TEXORA · LECTURE SHEET

Cost Behaviour and Terminology

How costs behave as activity changes — fixed, variable, semi-variable and step costs; the relevant range; separating mixed costs by the high–low method; and the specialist vocabulary of costing: sunk, opportunity, marginal, controllable and conversion costs.

AC-4205 · COST AND MANAGEMENT ACCOUNTING · TOPIC 2 · LEVEL 4 TERM 2

1. Cost behaviour patterns

Cost behaviour is the response of a cost to a change in activity (output, machine hours, labour hours). Every later technique in this course — break-even, flexible budgets, variance analysis, decisions — is cost behaviour applied.

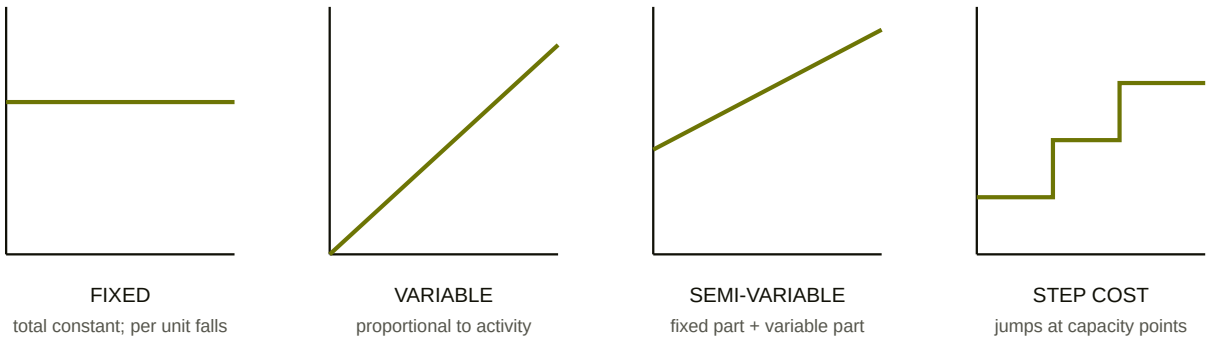


Figure 1 — The four behaviour patterns. Total fixed cost is flat while unit fixed cost falls with volume; variable cost is proportional; semi-variable cost starts above zero; step costs jump when a capacity threshold (e.g. one more supervisor, one more machine) is crossed.

PATTERN	BEHAVIOUR	TEXTILE EXAMPLES
Fixed	Total unchanged within the relevant range; per unit falls as output rises	Rent, insurance, managers' salaries, depreciation (straight-line)
Variable	Total proportional to activity; constant per unit	Fabric, dyes per kg, piece-rate wages, packing
Semi-variable (mixed)	A fixed charge plus a usage charge	Electricity (demand charge + units), machine maintenance, telephone
Step	Fixed over a band of activity, then jumps	Supervisors per shift, quality inspectors per line

The **relevant range** is the band of activity over which these patterns hold: push output beyond capacity and "fixed" costs step up (a second shift's supervision), while bulk discounts can bend "variable" costs. Behaviour statements are always *within the relevant range, in the short run*.

2. Separating a mixed cost — high–low method

Budgets and break-even analysis need mixed costs split into fixed and variable parts. The **high–low method** uses the highest and lowest activity observations:

$$\text{Variable cost per unit of activity} = (\text{Cost at high} - \text{Cost at low}) \div (\text{Activity at high} - \text{Activity at low})$$

WORKED EXAMPLE — SPLITTING A MAINTENANCE COST

A maintenance department's records: at 12,000 machine hours the cost was Tk 2,60,000; at 8,000 machine hours it was Tk 2,00,000.

Variable rate = (2,60,000 − 2,00,000) ÷ (12,000 − 8,000) = 60,000 ÷ 4,000 = **Tk 15 per machine hour**.

Fixed part = 2,00,000 − (8,000 × 15) = 2,00,000 − 1,20,000 = **Tk 80,000**.

Cost equation: **Cost = 80,000 + 15 × hours**. Check at the high point: 80,000 + 15 × 12,000 = 2,60,000 ✓.

Budget for 10,000 hours = 80,000 + 1,50,000 = **Tk 2,30,000**.

(The method is quick but uses only two observations — scatter-graph and regression methods use all the data and are preferred where accuracy matters.)

3. The terminology that follows from behaviour

TERM	MEANING	USE
Marginal cost	The extra cost of one more unit (in practice, the variable cost per unit in the short run)	Pricing and decisions, Topics 5 and 8
Sunk cost	Past cost, irrecoverable whatever is decided	Excluded from decisions
Opportunity cost	The benefit forgone by choosing one course over the next best	Included in decisions though never in the ledger
Controllable / uncontrollable	Whether a given manager can influence the cost within the period	Responsibility accounting and fair appraisal
Committed / discretionary fixed	Locked in by past decisions (depreciation, rent) vs set afresh each period (training, advertising)	Explains what cost-cutting can and cannot reach quickly

Conversion cost	Direct labour + factory overhead	Wet-processing cost analysis
Period vs product cost	Charged to the period's profit vs carried in stock until sale	Links cost accounting to financial statements (Topic 9)
Relevant cost	A future cost that differs between alternatives	The only costs that matter in a decision (Topic 8)

Glossary

Cost behaviour — how a cost responds to activity change. **Relevant range** — the activity band where assumed patterns hold. **High–low method** — splitting a mixed cost from its extreme observations. **Cost equation** — Total cost = Fixed + (Variable rate × Activity). **Marginal cost** — cost of one additional unit. **Sunk cost** — irrecoverable past cost. **Opportunity cost** — benefit forgone from the next-best use of a resource.

Key points

• Four patterns: fixed, variable, semi-variable, step — all valid only within the relevant range.
• High–low: variable rate Tk 15/hour and fixed Tk 80,000 from the worked data; budget at 10,000 h = Tk 2,30,000.
• Unit fixed cost falls with volume; total fixed cost does not — confusing the two is the classic costing error.
• Decisions use future, differing (relevant) costs; sunk costs never count, opportunity costs always do.

Practice

1. A factory's power cost behaves as: fixed Tk 60,000 per month plus Tk 15 per unit produced. Compute the total power cost and the power cost per unit at 8,000 units, and the per-unit figure at 12,000 units. What does the change illustrate?
2. Classify by behaviour: (a) dye per kg of fabric, (b) factory insurance, (c) electricity bill, (d) one supervisor per 25 machines, (e) carton cost per exported carton.
3. Using the high–low method on: 5,000 units → Tk 1,75,000; 9,000 units → Tk 2,55,000, find the variable cost per unit, the fixed cost, and the cost equation.
4. "The machine cost Tk 8,00,000 last year, so we must keep using it to recover the money." Identify the fallacy using the correct term.

Answer key

1. Total = 60,000 + 15 × 8,000 = **Tk 1,80,000**; per unit = 1,80,000 ÷ 8,000 = **Tk 22.50**. At 12,000 units: total = 60,000 + 1,80,000 = Tk 2,40,000; per unit = **Tk 20.00**. Illustration: the fixed element spreads over more units, so unit cost falls while the variable rate per unit stays Tk 15.
2. (a) Variable. (b) Fixed. (c) Semi-variable. (d) Step cost. (e) Variable.
3. Variable = (2,55,000 − 1,75,000) ÷ (9,000 − 5,000) = 80,000 ÷ 4,000 = **Tk 20 per unit**. Fixed = 1,75,000 − 5,000 × 20 = **Tk 75,000**. Equation: **Cost = 75,000 + 20 × units**. (Check at 9,000: 75,000 + 1,80,000 = 2,55,000 ✓.)
4. The Tk 8,00,000 is a **sunk cost** — spent and irrecoverable under every alternative. The keep/replace decision must compare only future costs and benefits of each option; trying to "recover" a sunk cost by continuing a loss-making use adds new losses to an old one.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib